

Mirza Farhan Bin Tarek

✉ mirzafarhanbin@gmail.com 🌐 mirzafarhan7.github.io 📄 mirzafarhanbintarek 📧 mfarhan007

Ph.D. Candidate in Computer Science — Large Language Models • Reinforcement Learning • AI Safety

Summary

Ph.D. candidate in Computer Science specializing in large language models, reinforcement learning, and AI safety for reliable clinical decision-making. Research focuses on hallucination mitigation, reward modeling, model alignment, and fairness-aware healthcare machine learning. Experienced in developing, evaluating, and deploying scalable ML and LLM pipelines using real-world clinical data, with peer-reviewed publications in ACM FAccT, IEEE/ACM CHASE, and ACM-BCB.

Education

University of Delaware, Newark, DE 2022 – 2027 (Expected)

Ph.D. in Computer Science; M.Sc. in Computer Science (Completed 2024)

Research: Hallucination mitigation in clinical LLMs.

Relevant Coursework: Machine Learning, Advanced Deep Learning, Database Systems

University of Staffordshire, Stoke-on-Trent, UK 2017 – 2018

Individual Modules, Erasmus+ KA1 Mobility Program

Relevant Coursework: Research Methods

United International University, Dhaka, Bangladesh 2014 – 2020

B.Sc. in Computer Science and Engineering, *Summa Cum Laude*

Relevant Coursework: Machine Learning, Probability and Statistics, Data Mining

Teaching Experience

Instructor & Graduate Teaching Assistant, University of Delaware 2022 – Present

- Instructor for CISC181: Introduction to Computer Science II, teaching Java programming and core object-oriented programming concepts, including classes, inheritance, polymorphism, interfaces, arrays, and collections.
- Designed assignments, exams, and programming projects to reinforce software development and problem-solving skills.
- Held regular office hours, mentored students on programming assignments, and provided individualized academic support.
- Assisted with undergraduate and graduate courses as a TA, including Data Mining and Machine Organization, supporting lectures, grading assignments, and facilitating student learning.

Lecturer, Brac University, Dhaka, Bangladesh 2019 – 2022

- Taught undergraduate courses CSE340: Computer Architecture and CSE461: Introduction to Robotics.
- Delivered lectures, designed course materials, and developed exams and assignments aligned with course learning objectives.
- Supervised student projects and provided mentorship on programming and hardware-related coursework.
- Served as Course Coordinator for CSE340, organizing examinations, coordinating with instructors and teaching assistants, and updating course materials to align with Outcome-Based Education (OBE) standards.

Research Experience

Graduate Researcher, University of Delaware 2022 – Present

- Conduct research on machine learning, reinforcement learning, and large language models in clinical decision systems.
- Developed methods to improve model reliability, fairness, and safety in healthcare AI applications.
- Published research at conferences, including IEEE/ACM CHASE '25, ACM-BCB '25, and ACM FAccT '23.

Academic Service

- Conference Reviewer2026
- Reviewed submissions for CHIL '25, CHIL '26, AISTATS '26, NeurIPS '26 Ethics & Data Track, MLHC '26, and BHI '26 Abstracts.

Selected Projects

- TA Dashboard Automation System — FastAPI, Python, MySQL2023 – 2024
- Designed and deployed a FastAPI-based analytics dashboard for multiple users; automated TA allocation and generated summary statistics and visual reports from MySQL data.
- Development of Machine Learning Models and a Product Predictor to Identify Patients at Risk for Diabetic Retinopathy in a Veteran Population — PyTorch, Transformers2023 – 2025
- Developed a machine learning (ML) model to classify patients into low- and high-risk categories for DR, allowing personalized screening intervals to optimize resource use and focus on high-risk individuals.
 - Applied statistical modeling and data visualization to analyze model fairness metrics (SPD, AOD) and interpret large-scale datasets.
 - Analyzed and modeled 20K+ patient records using XGBoost and Random Forest to predict diabetic retinopathy risk; achieved AUROC > 0.8 and deployed interpretable model outputs for clinical decision support.

Technical Skills

- Programming:Python, Java, C
- ML / Deep Learning:PyTorch, scikit-learn, Transformers, Hugging Face, NumPy, SciPy
- LLMs / AI Safety:Reward Modeling, Reinforcement Learning, RLHF/RLVR, Hallucination Evaluation, Model Alignment
- Data Analysis:Pandas, Matplotlib, Seaborn, Statistical Analysis, Hypothesis Testing
- Systems & Tools:Git, Docker, Linux, SLURM, Jupyter, LaTeX
- Databases & Web:MySQL, Oracle, REST APIs, FastAPI, HTML/CSS, JavaScript

Selected Publications

- Tarek Bin, Mirza Farhan,** and Rahmatollah Beheshti. “Reward Hacking Mitigation using Verifiable Composite Rewards.” arXiv preprint arXiv:2509.15557 (2025). *Accepted at ACM-BCB’25*.
- M. F. Bin Tarek,** R. Poulain, and R. Beheshti, “Fairness-Optimized Synthetic EHR Generation for Arbitrary Downstream Predictive Tasks,” 2025 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE), New York, NY, USA, 2025, pp. 234-244.
- Poulain, R., **Tarek, M. F. B.,** and Beheshti, R. (2023). Improving Fairness in AI Models on Electronic Health Records: The Case for Federated Learning Methods. FAccT ’23: Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency. Association for Computing Machinery, 2023, 1599–1608.

Awards

- Awarded NSF Student Travel Award for attending IEEE/ACM Chase’25
- Selected to participate in the Erasmus+ KA1 Mobility Program (2017)
- Completed Coursera Deep Learning Specialization (5-course series), Stanford University (2020)